20261012 Quiz

for (MATH210) Applied Complex Variables

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Exercise 36.6. Show that if $z \neq 0$ and a is a real number, then

$$|z^a| = \exp(a \ln |z|) = |z|^a,$$

where the principal value of $|z|^a$ is to be taken.

Solution. For $z \neq 0$, choose any value

$$L = \ln |z| + i(\operatorname{Arg} z + 2k\pi), \qquad k \in \mathbb{Z},$$

of $\log z$. The corresponding value $w = e^{aL}$ of z^a satisfies

$$|w| = e^{a \ln |z|} \left| e^{ia(\operatorname{Arg} z + 2k\pi)} \right| = e^{a \ln |z|},$$

since $a \in \mathbb{R}$. This modulus is independent of the chosen value of $\log z$. Since $|z| > 0$, its ordinary real power is

$$|z|^a = \exp(a \ln |z|).$$

Thus

$$\boxed{|z^a| = \exp(a \ln |z|) = |z|^a}.$$

□

Exercise 36.8. Let c, c_1, c_2, z be complex numbers with $z \neq 0$. Prove that, if all the powers involved are principal values, then

(a) $z^{c_1} z^{c_2} = z^{c_1 + c_2}$;

(b) $\dfrac{z^{c_1}}{z^{c_2}} = z^{c_1 - c_2}$;

(c) $(z^c)^n = z^{cn}$, where $n = 1, 2, \ldots$.

Solution. Use the principal-value convention $z^w = \exp(w \operatorname{Log} z)$ throughout this exercise.

(a) By the exponential addition law,

$$z^{c_1} z^{c_2} = e^{c_1 \operatorname{Log} z} e^{c_2 \operatorname{Log} z} = e^{(c_1 + c_2) \operatorname{Log} z} = \boxed{z^{c_1 + c_2}}.$$

(b) The denominator is nonzero; hence

$$\frac{z^{c_1}}{z^{c_2}} = \frac{e^{c_1 \operatorname{Log} z}}{e^{c_2 \operatorname{Log} z}} = e^{(c_1 - c_2) \operatorname{Log} z} = \boxed{z^{c_1 - c_2}}.$$

(c) Since n is a positive integer, raising a complex number to its nth power has its ordinary single-valued meaning. Thus

$$(z^c)^n = (e^{c\,\mathrm{Log}\,z})^n = e^{nc\,\mathrm{Log}\,z} = \boxed{z^{cn}}.$$

The common choice of $\mathrm{Log}\,z$ is essential to these identities; arbitrary independent choices of the multivalued powers need not satisfy them.

□

Exercise 36.9. Assuming that $f'(z)$ exists, state the formula for the derivative of $c^{f(z)}$.

Solution. Assume $c \neq 0$ and fix one value L of $\log c$. For this fixed choice, Sec. 35 defines the corresponding power by

$$c^{f(z)} = \exp(f(z)L).$$

Thus the chain rule gives

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}z}c^{f(z)} = c^{f(z)}f'(z)\,L}.$$

This is the textbook formula $(c^{f(z)})' = c^{f(z)}f'(z)\log c$, where $\log c$ denotes the *same fixed value* L throughout. For principal values, choose $L = \mathrm{Log}\,c$.

□

Exercise 38.1. Give details in the derivation of expressions (2), Sec. 37, for the derivatives of $\sin z$ and $\cos z$.

Solution. By their definitions in Sec. 37,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \qquad \cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

Applying the exponential differentiation rule and the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}z}\sin z = \frac{ie^{iz} + ie^{-iz}}{2i} = \frac{e^{iz} + e^{-iz}}{2} = \boxed{\cos z},$$
$$\frac{\mathrm{d}}{\mathrm{d}z}\cos z = \frac{ie^{iz} - ie^{-iz}}{2} = -\frac{e^{iz} - e^{-iz}}{2i} = \boxed{-\sin z}.$$

□

Exercise 38.4. Verify the identity $\sin^2 z + \cos^2 z = 1$ using

(a) identity (6) and relations (3) in Sec. 37;

(b) the lemma in Sec. 28 and the fact that the entire function $f(z) = \sin^2 z + \cos^2 z - 1$ has zero values along the real axis.

Solution.

(a) Setting $z_1 = z$ and $z_2 = -z$ in the addition formula

$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$$

and using $\cos(-z) = \cos z$, $\sin(-z) = -\sin z$, we get

$$1 = \cos 0 = \cos^2 z + \sin^2 z.$$

(b) The function $f(z) = \sin^2 z + \cos^2 z - 1$ is entire. For every real x, the real trigonometric identity gives $f(x) = 0$. Since the real axis contains a sequence of distinct zeros converging to, for example, 0, the lemma in Sec. 28 implies $f \equiv 0$ throughout $\mathbb{C}$. Thus $\sin^2 z + \cos^2 z = 1$ for all $z \in \mathbb{C}$.

□

Exercise 38.15. Find all roots of $\sin z = \cosh 4$ by equating the real parts and then the imaginary parts of $\sin z$ and $\cosh 4$.

Solution. Write $z = x + iy$, with $x, y \in \mathbb{R}$. Formula (13), Sec. 37, gives

$$\sin z = \sin x \cosh y + i \cos x \sinh y.$$

Consequently,

$$\sin x \cosh y = \cosh 4, \qquad \cos x \sinh y = 0.$$

If $y = 0$, the first equation becomes $\sin x = \cosh 4 > 1$, which is impossible. Hence $\cos x = 0$, so $x = \pi/2 + n\pi$. The first equation now reads $(-1)^n \cosh y = \cosh 4 > 0$, forcing n to be even and $\cosh y = \cosh 4$. Since $\cosh y$ is even and strictly increasing for $y > 0$, $y = \pm 4$. All roots are therefore

$$\boxed{z = \frac{\pi}{2} + 2n\pi \pm 4i \qquad (n \in \mathbb{Z})}.$$

□

Exercise 39.1. Verify that the derivatives of $\sinh z$ and $\cosh z$ are as stated in equations (2), Sec. 39.

Solution. From the definitions

$$\sinh z = \frac{e^z - e^{-z}}{2}, \qquad \cosh z = \frac{e^z + e^{-z}}{2},$$

we obtain

$$\frac{\mathrm{d}}{\mathrm{d}z} \sinh z = \frac{e^z + e^{-z}}{2} = \boxed{\cosh z},$$
$$\frac{\mathrm{d}}{\mathrm{d}z} \cosh z = \frac{e^z - e^{-z}}{2} = \boxed{\sinh z}.$$

□

Exercise 39.8. Give details showing that the zeros of $\sinh z$ and $\cosh z$ are as in the theorem in Sec. 39.

Solution. The identities from Sec. 39 yield

$$\sinh z = -i \sin(iz), \qquad \cosh z = \cos(iz).$$

By the zero sets of sine and cosine in Sec. 38,

$$\sinh z = 0 \iff iz = n\pi \iff z = -n\pi i \qquad (n \in \mathbb{Z}),$$

which is exactly the set $z = n\pi i$, since n ranges over all integers. Similarly,

$$\cosh z = 0 \iff iz = \frac{\pi}{2} + n\pi \iff z = -\left(\frac{\pi}{2} + n\pi\right) i.$$

Reindexing with $m = -n - 1$ gives the equivalent form

$$\boxed{z = \left(\frac{\pi}{2} + m\pi\right) i \qquad (m \in \mathbb{Z})}.$$

Conversely, the displayed identities show directly that every listed point is a zero, and no others are possible.

□

Exercise 39.16. By using one of the identities (9) and (10) in Sec. 39 and then proceeding as in Exercise 15, Sec. 38, find all roots of

(a) $\sinh z = i$;

(b) $\cosh z = \dfrac{1}{2}$.

Solution. Write $z = x + iy$, where $x, y \in \mathbb{R}$. From Sec. 39,

$$\sinh z = \sinh x \cos y + i \cosh x \sin y,$$
$$\cosh z = \cosh x \cos y + i \sinh x \sin y.$$

(a) The equation $\sinh z = i$ becomes

$$\sinh x \cos y = 0, \qquad \cosh x \sin y = 1.$$

Since $\cosh x \geq 1$, if $\cos y = 0$, then $\sin y = \pm 1$; the second equation requires $\sin y = 1$ and $\cosh x = 1$, hence $x = 0$. If instead $\sinh x = 0$, we again have $x = 0$, and the second equation gives $\sin y = 1$. Thus every root is

$$\boxed{z = \left(\frac{\pi}{2} + 2n\pi\right) i \qquad (n \in \mathbb{Z})}.$$

(b) The equation $\cosh z = 1/2$ gives

$$\cosh x \cos y = \frac{1}{2}, \qquad \sinh x \sin y = 0.$$

If $\sin y = 0$, then $\cos y = \pm 1$, and the first equation would require $\cosh x = 1/2$ or $-1/2$, both impossible. Therefore $\sinh x = 0$, i.e. $x = 0$, and $\cos y = 1/2$. It follows that

$$\boxed{z = \left(2n\pi \pm \frac{\pi}{3}\right) i \qquad (n \in \mathbb{Z})}.$$

□

Exercise 40.2. Solve the equation $\sin z = 2$ for z by

(a) equating real parts and then imaginary parts in that equation;

(b) using expression (2), Sec. 40, for $\sin^{-1} z$.

Solution.

(a) Let $z = x + iy$. Equating components of $\sin z = \sin x \cosh y + i \cos x \sinh y = 2$ gives

$$\sin x \cosh y = 2, \qquad \cos x \sinh y = 0.$$

If $y = 0$, then $\sin x = 2$, which is impossible. Thus $\cos x = 0$, so $x = \pi/2 + n\pi$. Positivity of 2 forces n to be even, and $\cosh y = 2$. Solving $e^y + e^{-y} = 4$ yields

$$y = \pm \ln(2 + \sqrt{3}).$$

(b) Expression (2), Sec. 40, gives the multivalued inverse formula

$$\sin^{-1} 2 = -i \log\bigl(2i + (1-4)^{1/2}\bigr) = -i \log\bigl(i(2 \pm \sqrt{3})\bigr).$$

Both $2 \pm \sqrt{3}$ are positive, so all logarithmic values are

$$\ln(2 \pm \sqrt{3}) + i\left(\frac{\pi}{2} + 2n\pi\right), \qquad n \in \mathbb{Z}.$$

Multiplication by $-i$, together with $\ln(2 - \sqrt{3}) = -\ln(2 + \sqrt{3})$, gives the same solutions as in part (a).

Therefore

$$\boxed{z = \frac{\pi}{2} + 2n\pi \pm i \ln(2 + \sqrt{3}) \qquad (n \in \mathbb{Z})}.$$

□

Exercise 40.3. Solve the equation $\cos z = 2$ for z.

Solution. For $z = x + iy$, use the identity

$$\cos z = \cos x \cosh y - i \sin x \sinh y.$$

The equation $\cos z = 2$ is therefore equivalent to

$$\cos x \cosh y = 2, \qquad \sin x \sinh y = 0.$$

The case $y = 0$ is impossible because then $\cos x = 2$. Hence $\sin x = 0$, so $x = n\pi$. As $\cosh y > 0$, the real-part equation forces n to be even and $\cosh y = 2$. Thus

$$\boxed{z = 2n\pi \pm i \ln(2 + \sqrt{3}) \qquad (n \in \mathbb{Z})}.$$

□

Exercise 42.2. Evaluate the following integrals:

(a) $\displaystyle\int_0^1 (1 + it)^2 \, dt$;

(b) $\displaystyle\int_1^2 \left(\frac{1}{t} - i\right)^2 dt$;

(c) $\displaystyle\int_0^{\pi/6} e^{i2t} \, dt$;

(d) $\displaystyle\int_0^{\infty} e^{-zt} \, dt$, where $\operatorname{Re} z > 0$.

Solution. The integrals of complex-valued functions of a real variable are evaluated componentwise (Sec. 42).

(a) Expanding the integrand,

$$\int_0^1 (1 + it)^2 \, dt = \int_0^1 (1 - t^2 + 2it) \, dt = \boxed{\frac{2}{3} + i}.$$

(b) Since $(t^{-1} - i)^2 = t^{-2} - 1 - 2i/t$,

$$\begin{aligned}\int_1^2 \left(\frac{1}{t} - i\right)^2 dt &= \left(-\frac{1}{t} - t - 2i \ln t\right)\Big|_1^2 \\ &= \boxed{-\frac{1}{2} - i \ln 4}.\end{aligned}$$

(c) Using $\frac{\mathrm{d}}{\mathrm{d}t}e^{2it} = 2ie^{2it}$,

$$\int_0^{\pi/6} e^{2it}\,\mathrm{d}t = \frac{e^{i\pi/3}-1}{2i} = \frac{\frac{1}{2}+i\frac{\sqrt{3}}{2}-1}{2i}$$
$$= \boxed{\frac{\sqrt{3}}{4}+\frac{i}{4}}.$$

(d) Write $z = x + iy$ with $x > 0$. Then

$$\left|e^{-zT}\right| = e^{-xT} \longrightarrow 0 \qquad (T \to \infty).$$

Since $z \neq 0$, the improper integral converges and

$$\int_0^{\infty} e^{-zt}\,\mathrm{d}t = \lim_{T\to\infty}\frac{1-e^{-zT}}{z} = \boxed{\frac{1}{z}}.$$

□

Exercise 42.3. Show that if m and n are integers, then

$$\int_0^{2\pi} e^{im\theta}e^{-in\theta}\,\mathrm{d}\theta = \begin{cases} 0 & \text{when } m \neq n, \\ 2\pi & \text{when } m = n. \end{cases}$$

Solution. Put $k = m - n \in \mathbb{Z}$. If $k = 0$, the integrand is 1, so the integral is 2π. If $k \neq 0$, then

$$\int_0^{2\pi} e^{ik\theta}\,\mathrm{d}\theta = \frac{e^{2\pi ik}-1}{ik} = 0,$$

because $e^{2\pi ik} = 1$ for every integer k. This proves both cases.

□

Exercise 43.4. Verify expression (14), Sec. 43, for the derivative of $Z(\tau) = z(\phi(\tau))$.
Suggestion. Write $Z(\tau) = x(\phi(\tau)) + iy(\phi(\tau))$ and apply the chain rule for real-valued functions of a real variable.

Solution. Write $z(t) = x(t) + iy(t)$ and $t = \phi(\tau)$. By the chain rule in real calculus,

$$\begin{aligned} Z'(\tau) &= \frac{\mathrm{d}}{\mathrm{d}\tau}x(\phi(\tau)) + i\frac{\mathrm{d}}{\mathrm{d}\tau}y(\phi(\tau)) \\ &= x'(\phi(\tau))\phi'(\tau) + iy'(\phi(\tau))\phi'(\tau) \\ &= (x'(\phi(\tau)) + iy'(\phi(\tau)))\phi'(\tau). \end{aligned}$$

Therefore

$$\boxed{Z'(\tau) = z'(\phi(\tau))\phi'(\tau)},$$

which is expression (14) in Sec. 43.

□

Exercise 43.5. Suppose that a function $f(z)$ is analytic at a point $z_0 = z(t_0)$ lying on a smooth arc $z = z(t)$ $(a \le t \le b)$. Show that if $w(t) = f(z(t))$, then

$$w'(t) = f'(z(t))z'(t)$$

when $t = t_0$.
Suggestion. Write $f(z) = u(x, y) + iv(x, y)$ and $z(t) = x(t) + iy(t)$, so that

$$w(t) = u(x(t), y(t)) + iv(x(t), y(t)).$$

Then apply the chain rule in calculus for functions of two real variables to write

$$w' = (u_x x' + u_y y') + i(v_x x' + v_y y'),$$

and use the Cauchy–Riemann equations.

Solution. At $t = t_0$, the real-variable chain rule gives

$$\begin{aligned} w'(t) &= (u_x x' + u_y y') + i(v_x x' + v_y y') \\ &= (u_x + iv_x)x' + (u_y + iv_y)y', \end{aligned}$$

where the partial derivatives are evaluated at $(x(t_0), y(t_0))$ and x', y' at t_0. Since f is analytic at z_0, the Cauchy–Riemann equations imply

$$u_y = -v_x, \qquad v_y = u_x,$$

and therefore

$$u_y + iv_y = -v_x + iu_x = i(u_x + iv_x).$$

Moreover, $f'(z_0) = u_x + iv_x$. It follows that

$$\begin{aligned} w'(t_0) &= f'(z_0)(x'(t_0) + iy'(t_0)) \\ &= \boxed{f'(z(t_0))z'(t_0)}. \end{aligned}$$

□

Exercise 46.3. Let $f(z) = \pi \exp(\pi \overline{z})$, and let C be the boundary of the square with vertices $0, 1, 1 + i, i$, oriented counterclockwise. Evaluate $\int_C f(z)\, \mathrm{d}z$.

Solution. Parametrize the four directed sides, in order, by

$$\begin{aligned} &C_1 : \ z = t, && 0 \le t \le 1, \\ &C_2 : \ z = 1 + it, && 0 \le t \le 1, \\ &C_3 : \ z = 1 - t + i, && 0 \le t \le 1, \\ &C_4 : \ z = i(1 - t), && 0 \le t \le 1. \end{aligned}$$

Using the conjugate of each parametrization, the contour-integral definition in Sec. 44 gives

$$\begin{aligned} \int_{C_1} f(z)\, \mathrm{d}z &= \pi \int_0^1 e^{\pi t}\, \mathrm{d}t = e^{\pi} - 1, \\ \int_{C_2} f(z)\, \mathrm{d}z &= i\pi e^{\pi} \int_0^1 e^{-i\pi t}\, \mathrm{d}t = 2e^{\pi}, \\ \int_{C_3} f(z)\, \mathrm{d}z &= \pi \int_0^1 e^{\pi(1-t)}\, \mathrm{d}t = e^{\pi} - 1, \\ \int_{C_4} f(z)\, \mathrm{d}z &= -i\pi \int_0^1 e^{-i\pi(1-t)}\, \mathrm{d}t = -2. \end{aligned}$$

Adding the four contributions,

$$\boxed{\int_C \pi e^{\pi \bar{z}}\, \mathrm{d}z = 4(e^{\pi} - 1)}.$$

□

Exercise 46.9. Let C denote the positively oriented unit circle $|z| = 1$ about the origin.

(a) Show that if $f(z)$ is the principal branch

$$z^{-3/4} = \exp\left(-\frac{3}{4}\operatorname{Log} z\right) \qquad (|z| > 0,\ -\pi < \operatorname{Arg} z < \pi)$$

of $z^{-3/4}$, then

$$\int_C f(z)\, \mathrm{d}z = 4\sqrt{2}\, i.$$

(b) Show that if $g(z)$ is the branch

$$z^{-3/4} = \exp\left(-\frac{3}{4}\log z\right) \qquad (|z| > 0,\ 0 < \arg z < 2\pi)$$

of the same power function, then

$$\int_C g(z)\, \mathrm{d}z = -4 + 4i.$$

This exercise demonstrates how the value of an integral of a power function depends in general on the branch that is used.

Solution. The branch in (a) is undefined at $z = -1$, and the branch in (b) is undefined at $z = 1$. Strictly speaking, neither integral over the entire circle is an ordinary contour integral as defined in Sec. 44. Following the intent of the exercise, we interpret the textbook notation as an improper integral along the circle with the cut point omitted, taking limits from the two sides.

(a) Parametrize the circle by $z = e^{i\theta}$, $-\pi < \theta < \pi$. On this arc, $\operatorname{Log} z = i\theta$ and $\mathrm{d}z = ie^{i\theta}\, \mathrm{d}\theta$. The intended integral is

$$\begin{aligned} I_f &:= \lim_{\varepsilon \to 0^+} \int_{-\pi+\varepsilon}^{\pi-\varepsilon} ie^{i\theta/4}\, \mathrm{d}\theta \\ &= 4(e^{i\pi/4} - e^{-i\pi/4}) = \boxed{4\sqrt{2}\, i}. \end{aligned}$$

(b) Let $h(z) = \ln|z| + i\theta$, $0 < \theta < 2\pi$, denote the single-valued logarithmic branch written as $\log z$ in the exercise. With $z = e^{i\theta}$ on this arc, $h(z) = i\theta$. Hence

$$\begin{aligned} I_g &:= \lim_{\varepsilon \to 0^+} \int_{\varepsilon}^{2\pi-\varepsilon} ie^{i\theta/4}\, \mathrm{d}\theta \\ &= 4(e^{i\pi/2} - 1) = \boxed{-4 + 4i}. \end{aligned}$$

These are the values denoted by $\int_C f(z)\, \mathrm{d}z$ and $\int_C g(z)\, \mathrm{d}z$, respectively, in the exercise. They differ because the branches use different argument intervals.

□

Exercise 46.10. With the aid of the result in Exercise 3, Sec. 42, evaluate

$$\int_C z^m \overline{z}^n \, \mathrm{d}z,$$

where $m, n \in \mathbb{Z}$ and C is the unit circle $|z| = 1$, oriented counterclockwise.

Solution. Use the parametrization $z = e^{i\theta}$, $0 \le \theta \le 2\pi$. On C,

$$\overline{z} = e^{-i\theta}, \qquad \mathrm{d}z = ie^{i\theta} \, \mathrm{d}\theta.$$

Thus

$$\begin{aligned} \int_C z^m \overline{z}^n \, \mathrm{d}z &= i \int_0^{2\pi} e^{im\theta} e^{-in\theta} e^{i\theta} \, \mathrm{d}\theta \\ &= i \int_0^{2\pi} e^{i(m+1)\theta} e^{-in\theta} \, \mathrm{d}\theta. \end{aligned}$$

By Exercise 42.3, the last integral is 2π when $m + 1 = n$ and 0 otherwise. Hence

$$\boxed{\int_C z^m \overline{z}^n \, \mathrm{d}z = \begin{cases} 2\pi i & \text{if } n = m + 1, \\ 0 & \text{if } n \neq m + 1. \end{cases}}$$

□